

SET 2

Time: 30 Minutes Total Marks: 50

STUDENT DETAILS (BLOCK LETTERS)

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SECTION A – PRACTICAL CODING (30 Marks)

Create ONE HTML file using on-page `<style>` and `<script>` only.

Include: Input field, Button, Output section.

Each step must use previous step result (chain logic).

TASK REQUIREMENTS:

1. Take a STRING input (only alphabet characters, length 3–10).
2. Convert string to uppercase.
3. Count total vowels and total consonants.
4. Generate Pattern 1 – Reverse Alphabet Ladder (height = string length).
5. Generate Pattern 2 – Hollow Square (size = vowel count + 3).
6. If vowel count > consonant count → Display VOWEL DOMINANT. Else → Display CONSONANT DOMINANT.

SAMPLE ILLUSTRATION

INPUT

Text = 'code'

PROCESS

Uppercase = CODE

Vowels = O, E $\rightarrow$ 2

Consonants = C, D $\rightarrow$ 2

Length = 4 $\rightarrow$ Ladder Height = 4

D

CD

BCD

ABCD

Square Size = Vowel Count + 3 = 5

* *

* *

* *

OUTPUT

CONSONANT DOMINANT (because vowels not greater than
consonants)

SECTION B – MCQ (20 Marks)

Q1. Which data structure uses FIFO?

- A) Stack
- B) Queue ✓
- C) Tree
- D) Graph

Q2. Foreign Key is used to:

- A) Delete rows
- B) Connect tables ✓
- C) Encrypt data
- D) Sort records

Q3. Time complexity of linear search is:

- A) $O(1)$
- B) $O(n)$ ✓
- C) $O(\log n)$
- D) $O(n \log n)$

Q4. Normalization reduces:

- A) Redundancy ✓
- B) Index size
- C) CPU speed
- D) Network load

Q5. Recursion requires:

- A) Loop only
- B) Base condition ✓
- C) SQL query
- D) API call

Q6. HTTP 200 indicates:

- A) Error
- B) Success ✓
- C) Redirect
- D) Timeout

Q7. API stands for:

- A) Application Programming Interface ✓
- B) Advanced Process Input

- C) Automated Program Index
- D) Application Private Internet

Q8. Git clone is used to:

- A) Upload code
- B) Download repository ☒
- C) Merge branch
- D) Delete branch

Q9. Cloud computing mainly provides:

- A) Physical CPU only
- B) On-demand resources ☒
- C) Printer sharing
- D) Offline storage

Q10. Multithreading improves:

- A) Single task delay
- B) Sequential execution
- C) Concurrent execution ☒
- D) Database size